

Course Project - Part 1

46770 Integrated Energy Grids

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INTRODUCTION

The given report was prepared and submitted as part of assessment in course 46770 Integrated Energy Grids. The report details 5 problems centred around energy system optimization. Tasks a)...d) were performed in Python programming language and are available in the delivered ZIP file.

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PROBLEM (A) - INITIAL GRID CREATION

1.1 Problem Definition

The country chosen for this project is Denmark, since it has a long history of using different renewable energy technologies. This provides a good foundation for finding and using historical data on renewable electricity production in Denmark Energinet [2026]. Combined with these historical data and the PyPSA tool, this creates a solid basis for determining the optimal capacities of the different generation technologies Authors [2024]. The main techno-economic assumptions of the modelled system are listed in Table 1.

Technology	Investment [€/MW]	Capital cost [€/MW/yr]	FOM [%]	VOM [€/MWh]	Fuel [€/MWh _{th}]	Marginal cost [€/MWh]	Efficiency [-]	Lifetime [years]
Solar	676570.3	63114.923985	1.7275	0.0106	0.000	0.010600	1.000	37.5
Onshore wind	1139882.6	107441.457021	1.2347	1.5080	0.000	1.508000	1.000	28.5
Offshore wind	1769117.1	184567.393550	2.3741	0.0212	0.000	0.021200	1.000	30.0
OCGT	470485.3	48739.697513	1.7784	4.7620	24.568	65.423728	0.405	25.0

Table 1. Techno-economic assumptions used in the model.

Table 1 summarises the trade-offs between the technologies included in the model: solar, onshore wind, offshore wind, and OCGT. The renewable technologies are characterised by very low marginal costs, while OCGT has the highest marginal cost due to fuel consumption. However, OCGT has the lowest investment cost and provides dispatchable generation, whereas the renewable technologies depend on weather conditions. Among the renewables, offshore wind has the highest investment and annual capital cost, while solar is the least expensive. Overall, renewables are economically attractive because of their low operating costs, but their variability means that a conventional backup technology such as OCGT is still needed to ensure system reliability.

The annual electricity mix is based on the techno-economic assumptions and shown in Figure 1. Here it is displayed that the main energy mix consists of onshore wind and OCGT, with a smaller contribution from solar. Offshore wind is not included, which is most likely due to the higher cost 1.

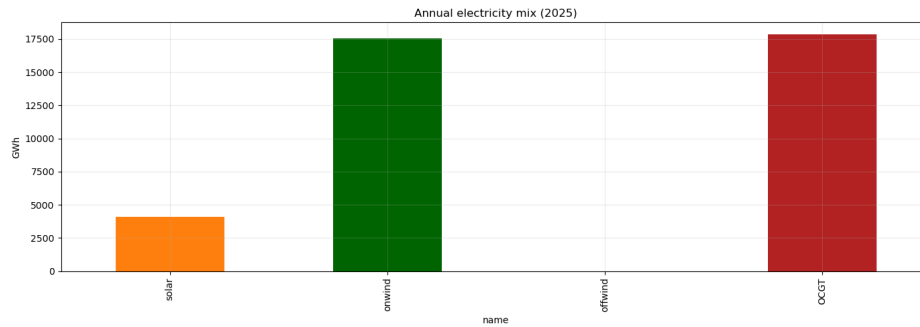


Figure 1. Annual electricity mix for Denmark in 2025.

Figures 2 and 3 display the difference in dispatch from the first week in July during summer to the first week in January during winter. In summer we have a nice mix of onshore wind, although solar mainly contributes during the day and onshore wind contributes at all hours. Limiting the need for support OCGT to mainly being outside midday.

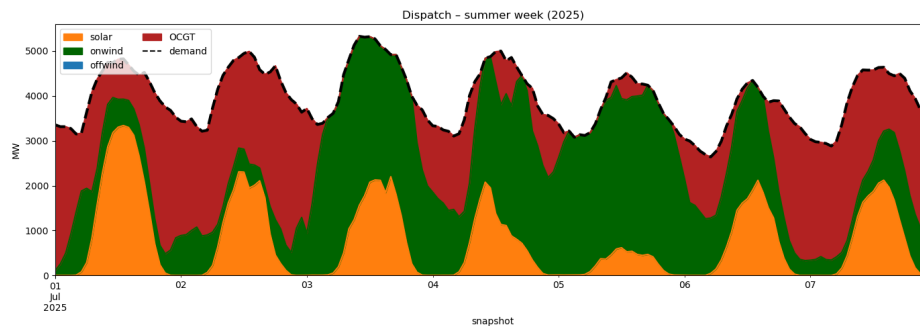


Figure 2. Dispatch for one week in summer, 01/07–07/07/2025.

In contrast there is little to no solar during the winter week and onshore wind being the biggest contributor to power. There are still times where it is necessary for OCGT to step in and support but there is less of a pattern as wind is less predictable. The results from the summer and winter dispatch indicated that onshore wind is the main contributor of power and due to its unreliability it is essential to have a flexible backup generator such as OCGT.

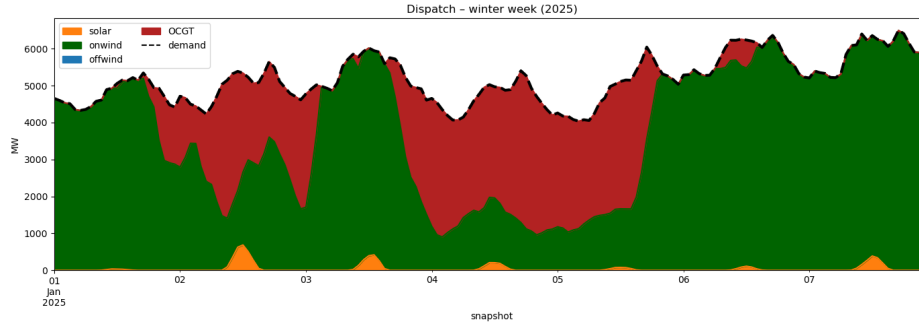
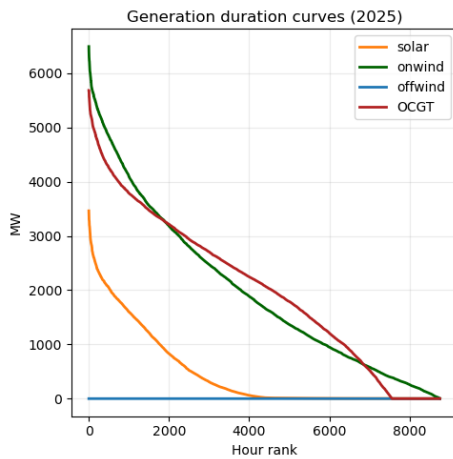
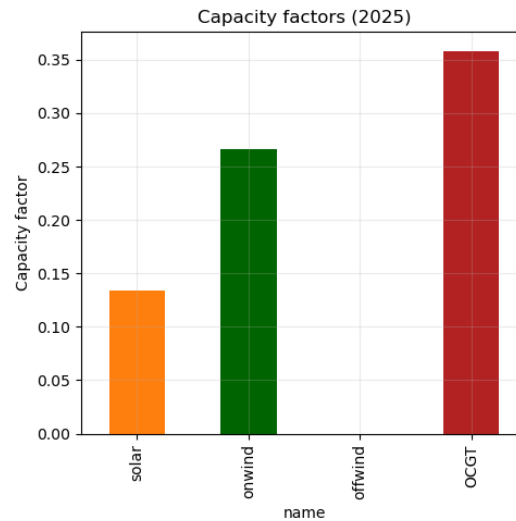


Figure 3. Dispatch for one week in winter, 01/01–07/01/2025.

The same pattern can also be seen by considering the duration curves and capacity factors shown in Figure 4. Onshore wind and OCGT contribute a lot of power across many hours, thereby being the key delivering components of the system. On the other hand solar contributes at approximately half of the number of hours and power, compared to the other technologies. In conclusion, onshore wind forms the main renewable backbone of the system, while OCGT provides balancing and security of supply.



(a) Generation duration curves for Denmark in 2025.



(b) Capacity factors for the technologies in Denmark in 2025.

Figure 4. Generation duration curves and capacity factors used to assess the contribution of different technologies.

PROBLEM (B) - INITIAL GRID SOLUTIONS DEPENDABILITY ON WEATHER YEARS

2.1 Problem Definition

The distribution of installed capacities found as the optimal solution is dependent on the forecasted production from wind and solar. To understand how prone the optimal capacities for different technologies are to changes in variable renewable energy production.

Production data from Energinet through Energi Data Service Energinet [2026], data from the years 2020 up to and including 2025 were used. In Figure 5 the change of suggested capacities each year is displayed against the average capacity for available technologies.

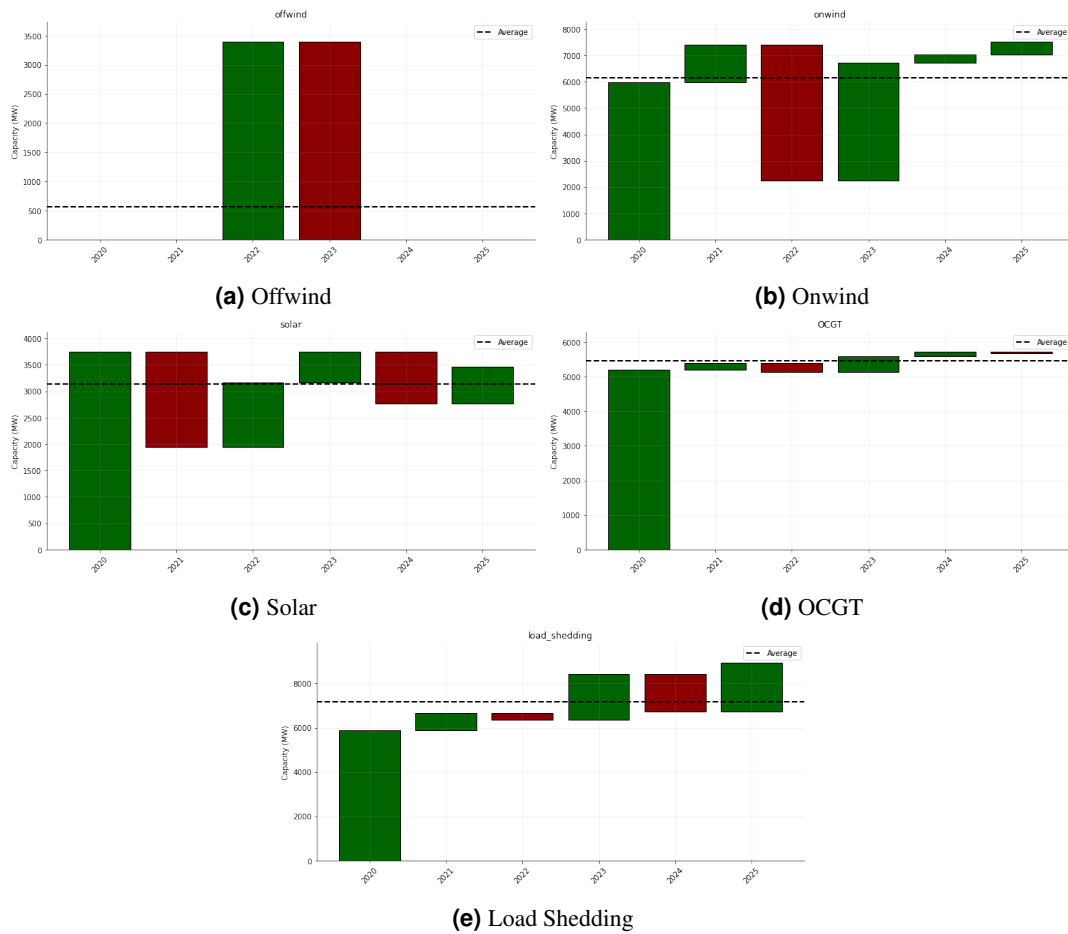


Figure 5. Average capacity and variability obtained for every generator using different weather years

Figure 5 clearly shows the results of relying on variable renewable energy technologies specifically solar, on- and offshore wind all of which are highly sensitive to weather conditions. These technologies indicate substantial year to year fluctuations, reflecting the differences in both seasonal pattern and total resource availability. In years with stronger availability of e.g. wind, the model favors increased deployment of onshore wind capacity, while in weaker wind years, the system shifts towards solar capacity or even more reliable technologies. The results indicate that the model actively adapts the technology mix to achieve the most favorable renewable resource energy in each of the weather years.

Offshore wind shows a distinct behavior compared to onshore wind and solar. It is only selected in the generation mix in one of the analyzed years, suggesting that its higher capital cost makes it competitive only under specific resource conditions. In 2022 it shows offshore wind offers significantly more favorable or stable production profile, stating its worth the in the generation mix. However, its absence in the other years, shows the sensitivity of both cost and interannual variability in wind resources offshore. The capacity of open cycle gas turbines remains, in contrast to renewable resources, relatively steady across all investigated years. It shows the importance of a dispatchable backup technology that compensates during low renewable generation periods. Since periods with low renewable resources occur in all years, the OCGT capacity is primarily determined by the need to ensure system reliability rather than by the variability in renewable resource availability. Resulting in minimal variability compared to variable renewable energy technologies. Load shedding follows a pattern similar to the variability of renewable technologies, increasing in years where renewable generation generally is less favorable. It shows the importance of accounting for interannual variability, as the system need reliability to always support the demand. Load shedding is practically seen as a solution, when no other options are present, which means it should be avoided when possible.

PROBLEM (C) - ADDING STORAGE CAPABILITIES TO THE GRID

3.1 Problem Description

Electricity systems with a high share of renewable energy sources are characterised by significant temporal variability in generation due to changing environmental conditions. In particular, wind and solar generation exhibit strong hourly, daily, and seasonal fluctuations, which are not necessarily aligned with demand profiles. This mismatch leads to two key system challenges: (i) increased reliance on dispatchable generators with higher marginal costs during periods of low renewable output, and (ii) curtailment of renewable generation during periods of surplus production.

Energy storage plays a central role in mitigating these effects by shifting energy across time. By storing excess renewable generation and releasing it during periods of scarcity, storage technologies reduce both curtailment and the need for expensive backup generation. The effectiveness of storage depends on its technical characteristics, including energy capacity (MWh), power capacity (MW), round-trip efficiency, and storage duration. In this study, two complementary storage technologies are introduced to capture different balancing timescales: **lithium-ion batteries** for short-term flexibility and **hydrogen-based storage** for long-term energy shifting.

3.2 Battery Storage

Lithium-ion battery storage is included to provide short-term balancing of intra-day fluctuations in renewable generation. Batteries are characterised by high round-trip efficiency and relatively low energy storage duration, making them suitable for smoothing short-term variability and reducing peak demand.

In the model, the battery is represented as a single storage unit with a fixed duration (energy-to-power ratio), meaning that the energy capacity is proportional to the installed power capacity. This simplifies the representation while capturing the essential operational behaviour of grid-scale battery systems. The battery storage parameters were sourced from PyPSA reference data Official combining of the indices `Lithium-Ion-NMC-bicharger` and `Lithium-Ion-NMC-store`.

3.3 Hydrogen Storage

Hydrogen storage is included to capture long-term and seasonal balancing capabilities. The system consists of three components: an electrolyser, a hydrogen storage unit, and a conversion technology (fuel cell or turbine) to generate electricity from stored hydrogen.

During periods of excess renewable generation, electricity is used to produce hydrogen via electrolysis. The hydrogen is stored and later converted back into electricity when generation is insufficient to meet demand. The system was modelled in three parts: electrolyser (`PEM electrolyzer` in PyPSA reference data Official), hydrogen storage (`hydrogen storage underground` in PyPSA reference data Official) and hydrogen turbine (`fuel cell` in PyPSA reference data Official).

3.4 System Performance

Primary system metric in a single-node optimisation is the resource capacities considering demand and cost. The performance of the system is evaluated based on the optimal capacity mix, dispatch patterns, and storage utilisation.

The annual electricity mix illustrates the contribution of each generation technology to total demand. A high share of renewable generation is expected, with OCGT providing backup during periods of low renewable availability.

Table 2. Comparison between optimised dispatch capacities in the model.

Resource	1st Model Capacities [MW]	2nd Model Capacities [MW]
Solar	2044.78	2327.57
Onshore wind	2861.25	2820.61
Offshore wind	713.16	758.73
Open cycle gas turbine	2803.19	2584.14
Battery	-	278.79
Hydrogen turbine	-	0

Winter Dispatch

During winter, higher demand and lower solar availability lead to increased reliance on wind generation and backup capacity. Storage systems play a critical role in balancing variability. Batteries exhibit frequent charge-discharge cycles on a daily basis. Hydrogen storage is not utilised at all due to zero capacities in the model.

State-of-charge profiles for both storage technologies provide insight into their operational roles and timescales.

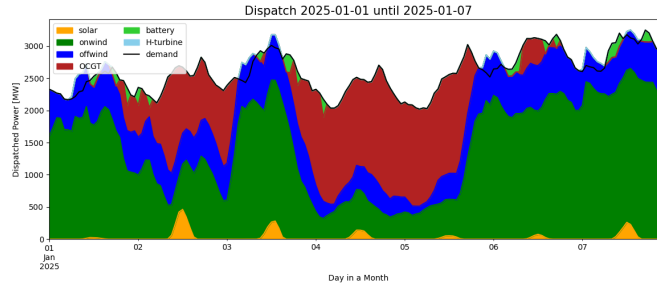


Figure 6. Energy dispatch of the resources during one week in the summer.

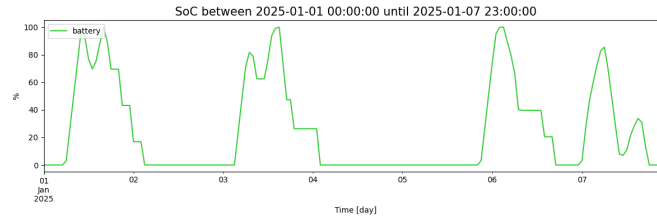


Figure 7. State of charge of the battery during one week in the summer.

Summer Dispatch

In summer, high solar generation leads to periods of surplus electricity, increasing the importance of storage. Batteries absorb intra-day solar peaks and release energy during evening demand. Hydrogen storage is not utilised again. The model has decided to still use considerable amount of OCGT which should not be the case in a real system and this behaviour should be further investigated later.

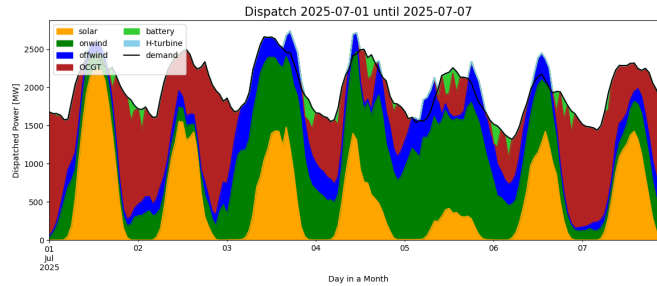


Figure 8. Energy dispatch of the resources during one week in the summer.

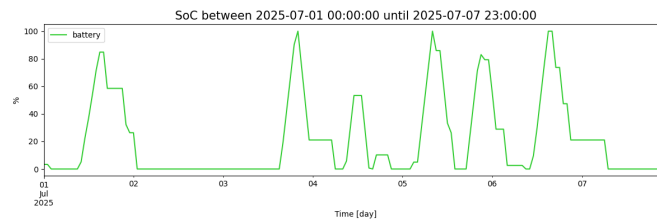


Figure 9. State of charge of the battery during one week in the summer.

Currently, there were significant differences in the system that was modelled by the author of step A and step C. This discrepancy is acknowledged and will be dealt with later. The current deliverable exemplifies the work progress and is not meant to represent the final product of work.

PROBLEM (D) - CONNECTING NEIGHBOURING COUNTRIES TO THE GRID

4.1 Problem Definition

The model is extended from a one-node model to a multi country model, by incorporating the neighbouring countries Germany, Sweden and Norway. The connection is established using HVAC lines with fixed interconnector capacities, and it is assumed that the generation capacities in the neighbouring countries are fixed. The interconnector capacity values for each country is fixed using data from Energinet system plan Energinet [2016]. The power flows are modeled using a linearised AC power flow formulation (DC approximation) with voltage level 400 kV and line reactance $x = 0.1$. The data used for the neighbouring countries are from the OPSD dataset introduced in the course, where Germany, Sweden and Norway load and demand data is available from 2015-2020. The year used in this task is 2019, with the OPSD data representing Denmark and the other chosen countries. Furthermore, we split the danish market into DK1 and DK2. The closed cycle in this task consists of

$$DK1 - DK2 - DE - DK1$$

4.2 System Performance

The multi country network shows that the optimum shifts toward Solar and onshore wind, where offshore-wind, OCGT and hydrogen are not selected. This is showcased in table 3.

Technology	DK1 [MW]	DK2 [MW]	Total [MW]
Solar PV	1111.99	3298.47	4410.46
Onshore wind	8850.14	5261.80	14111.94
Offshore wind	0.00	0.00	0.00
OCGT	0.00	0.00	0.00
Battery power	0.00	652.38	652.38
Hydrogen chain	0.00	0.00	0.00
Load shedding energy [MWh]	1912.66	1008.84	2921.50

Table 3. Optimized Danish capacities in the network model

This means that the model uses transmission across the border plus with a small DK2 battery as flexibility. Therefore, it is apparent that transmission to other countries count as a balancing strategy, which is able to partly substitute local backup and storage, as we see from the results. Since the hydrogen and OCGT is 0 with a modest battery for DK2, the balancing is optimized through the transmission to the other countries. We see from the line flows table 4 that congestion matters in this model, as the lines are fully loaded in a lot of the time steps.

Line	Capacity [MW]	Avg. flow [MW]	Hours at cap.	Avg. loading [%]
DK1-DK2	590	437.89	4528	74.22
DK1-DE	2500	651.65	0	26.07
DK2-DE	585	522.19	6878	89.26
DK2-SE	1300	952.85	4269	73.30
DK1-NO	1632	1214.84	4609	74.44

Table 4. Line-Loading for the multi-country model

The table shows that the lines hit capacity 4 out of 5 times frequently as represented by the average loading. The only line that doesn't bind is DK1-DE which has an average loading of about 26%. The table shows that DK2 is a heavily constrained zone, as it operates with fully loaded lines in DK2-DE and DK2-SE a lot of the times. DK2 therefore has less freedom to rely on neighbouring transfers in the hours, which could be why DK2 has the battery in table 3.

PROBLEM (E) - CONFIRMING THE CALCULATIONS BY MANUAL MEANS

5.1 Problem Definition

This step focuses on the first time step of the optimized model in the previous task, where the line-flow is calculated by hand. We will use the bus injections from the first time step to help calculate the optimal power flowing through every line.

For the hand calculation, we consider only the electrical transmission network from task d. The buses are ordered as

$$[1, 2, 3, 4, 5] = [DK1, DK2, DE, SE, NO],$$

and the transmission lines are ordered as

$$[1, 2, 3, 4, 5] = [DK1-DK2, DK1-DE, DK2-DE, DK2-SE, DK1-NO].$$

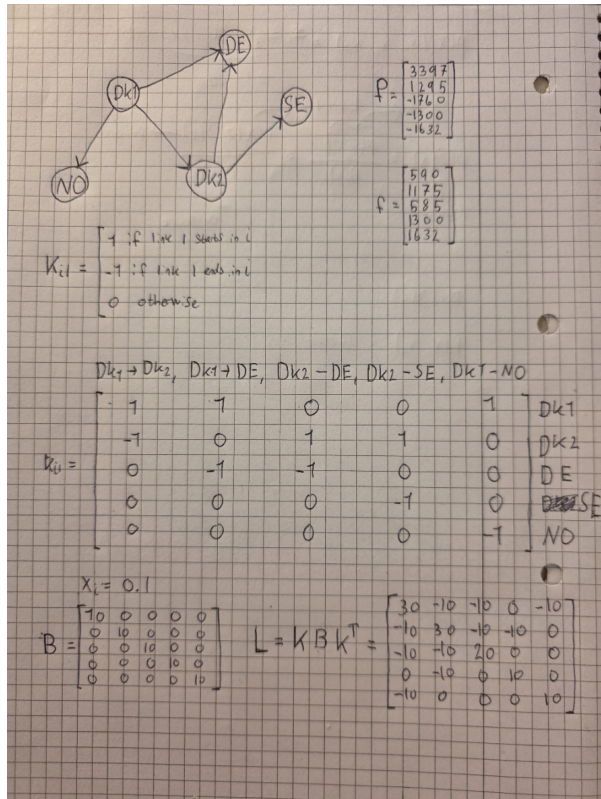
The line orientation is chosen to be consistent with the positive PyPSA flow direction.

$$DK1 \rightarrow DK2, \quad DK1 \rightarrow DE, \quad DK2 \rightarrow DE, \quad DK2 \rightarrow SE, \quad DK1 \rightarrow NO.$$

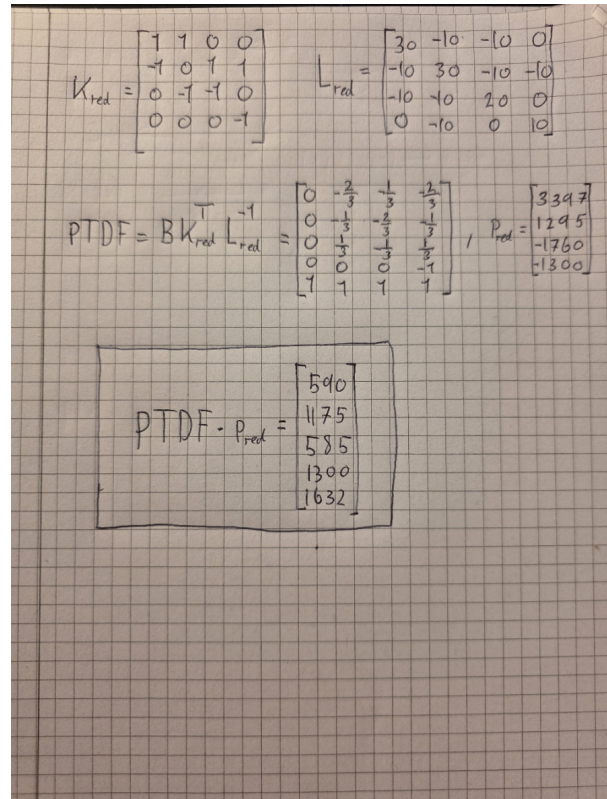
This ordering is used consistently in the handwritten derivation below for the incidence matrix, the weighted Laplacian matrix, the PTDF matrix, and the final line-flow calculation. The objective is to calculate the line flows from the first time step, and the nodal imbalance vector in the first time step is used, which is:

$$p = \begin{bmatrix} 3397 \\ 1295 \\ -1760 \\ -1300 \\ -1632 \end{bmatrix} \text{ MW}$$

5.2 Calculation



(a) Incidence Matrix and Weighted Laplacian Matrix



(b) PTDF matrix and Line flows

When having derived the Incidence Matrix from the graph, we use that together with the line reactances Matrix to calculate the Weighted Laplacian Matrix. Since the Laplacian is singular and therefore not invertible, we have to reduce it by introducing a slack bus, which in this case is the NO node and the DK1-NO line. By removing that row and column from the Incidence and Laplacian matrix, we are able to calculate the PTDF matrix, however, the line and node are not removed from the network itself though. We multiply the PTDF Matrix with the reduced line injections vector in order to obtain the line flows, which match the first time step of the optimized model perfectly, which is:

$$f = \begin{bmatrix} 590 \\ 1175 \\ 585 \\ 1300 \\ 1632 \end{bmatrix} \text{ MW},$$

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